Gideon A. Afriyie

Portfolio Website LinkedIn

Education

- **Gettysburg College** — Gettysburg, PA – USA
  B.A. Mathematics, Minor in Economics — *2023 – 2027*
  - Honors: AMS–Math Alliance F-GAP Scholar; Taylor Gaw '13 Endowed Fellowship; Seibert Project Award

Experience

- **Advancing Science Lab** — On-campus
  Research Assistant — *Aug 2026 – Present*
  - Integrated elevation, transportation, hydrography, and trail datasets into a tactile 3D model of South Mountain.
  - Applied SQL-style filtering and spatial clipping to extract 11 highway routes and selected waterways from GIS datasets.
  - Reprojected and aligned raster and vector layers in QGIS; exported processed datasets as GeoPackages.
  - Developed tactile map features using graduated road widths, recessed waterways, and a raised Appalachian Trail.
  - Produced an initial 3D-printed prototype; currently evaluating scale, vertical exaggeration, and feature clarity next print.

- **Under Canvas Inc.**
  Data Analytics Fellow — *May – Jul 2025*
  - Analyzed a $1.6M compensation dataset using Python and Excel to identify anomalies across employee groups.
  - Developed quantitative recommendations for incentive restructuring; findings presented to leadership.

Publications

- **Afriyie, Gideon A.** (2026). Extreme Value Theory Analysis of Prime Gap Distributions: Statistical Analysis of Cramér's Conjecture and Light-Tailed Behavior. *Statistics & Probability Letters*. DOI: 10.1016/j.spl.2026.110898
- Haque, R., Kebede, B., **Afriyie, G. A.**, & Yusuf, H. (2026). Predicting Urban Air Quality Trends Using Machine Learning and Public Satellite/Sensor Data. *Summer Proceedings 2026*, Chapter 16. Link

Projects

- **Quantitative Risk & Portfolio Optimization** — GitHub
  - Analyzed 6+ years of prices across 8 assets, estimating returns, volatility, and correlations (VOO–VIXY: -0.754).
  - Optimized portfolios using mean-variance analysis and efficient frontiers, applying KKT to restrict short selling.

Skills

Python, Excel, SQL, LaTeX

Research & Programs

Probability for AI — Stanford University — Oct–Nov 2026
TPU Research Cloud — Google — May 2026
AI Summer School — MIT/Harvard IAIFI — August 2026